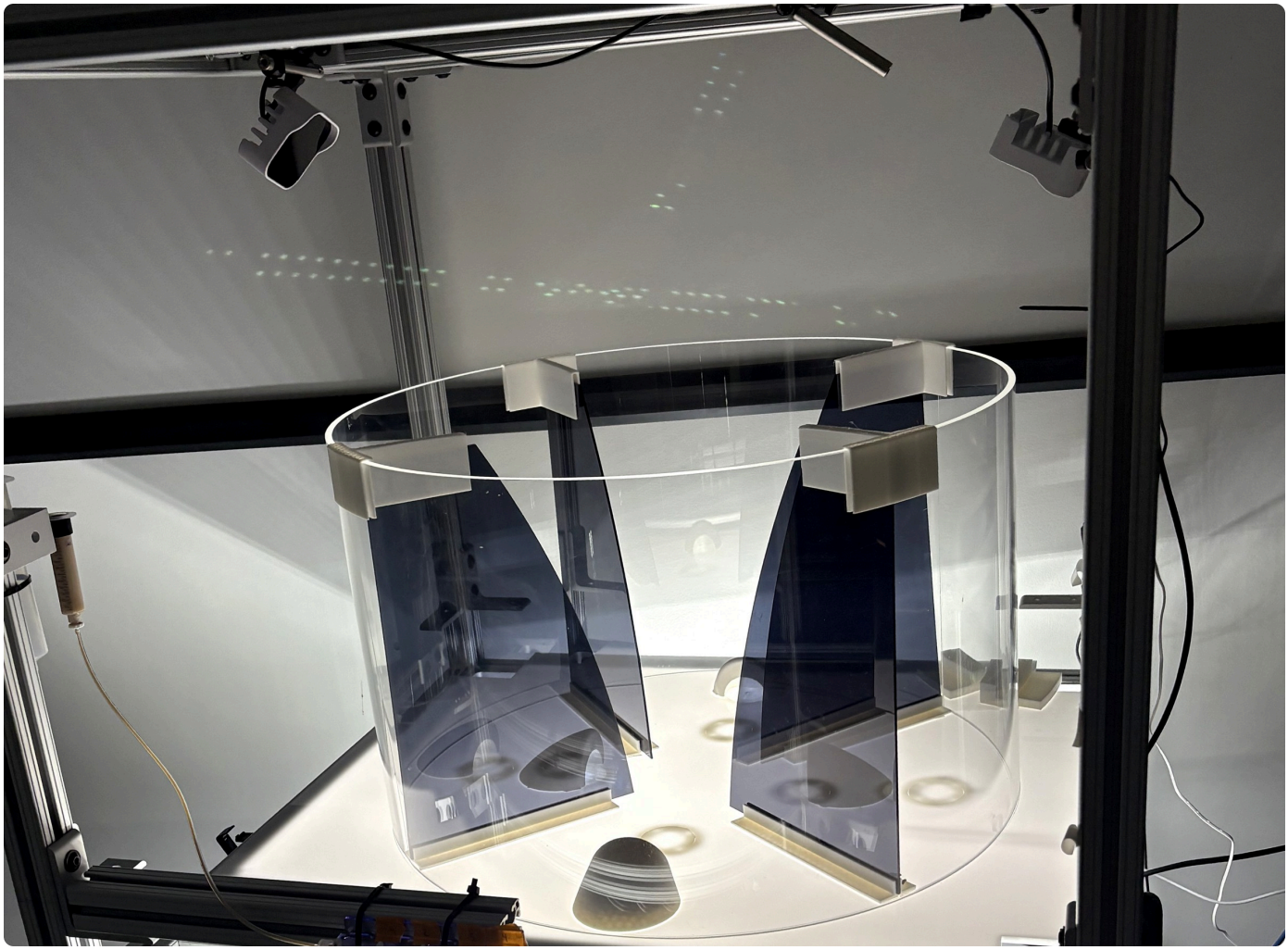
Bora Mandic

Mechanical Design Portfolio · Stanford University · boram@stanford.edu



Behavioral rig I designed and built: aluminum-extrusion frame, floor ports, and arena wall holders.

ABOUT

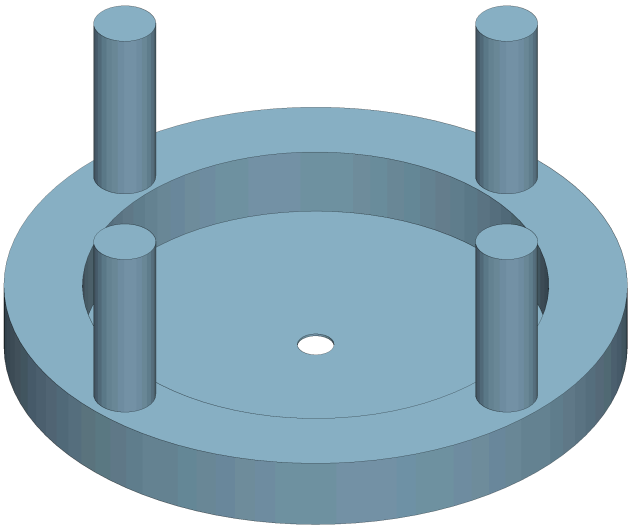
I'm a Stanford mechanical engineering student designing hardware for in-vivo mouse neuroscience: a full behavioral rig, head fixation, and surgery and histology fixtures. Every design here is parametric in Fusion, so parts can be quickly adjusted per experiment or per mouse, and each was printed, tested, and refined in the lab.

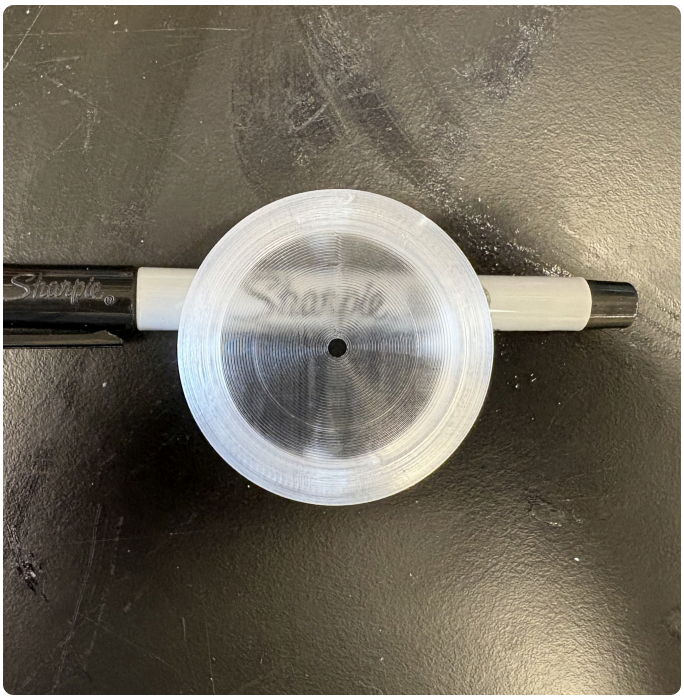
SELECTED WORK

- Behavioral rig: frame, arena port system (3 design generations, 24 versions), wall holders
- Circular head mount: 9 versions
- Lab fixtures: plating aid, surgery mount holder

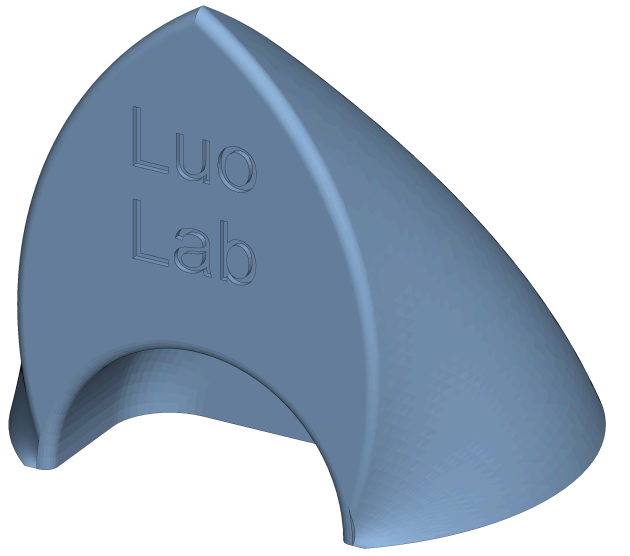
TOOLS

Autodesk Fusion (parametric CAD) · FDM printing (Bambu, Prusa) · laser cutting · Arduino · MATLAB

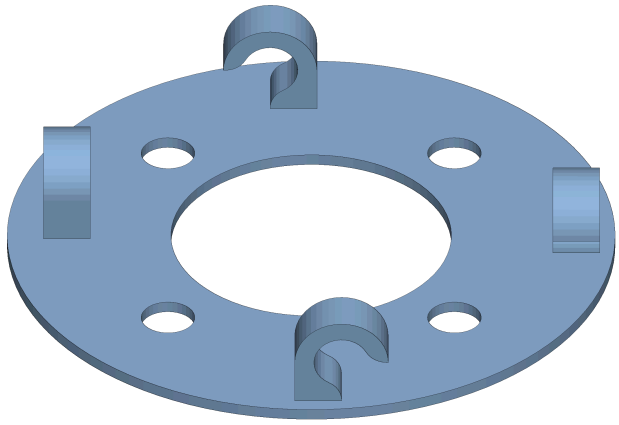




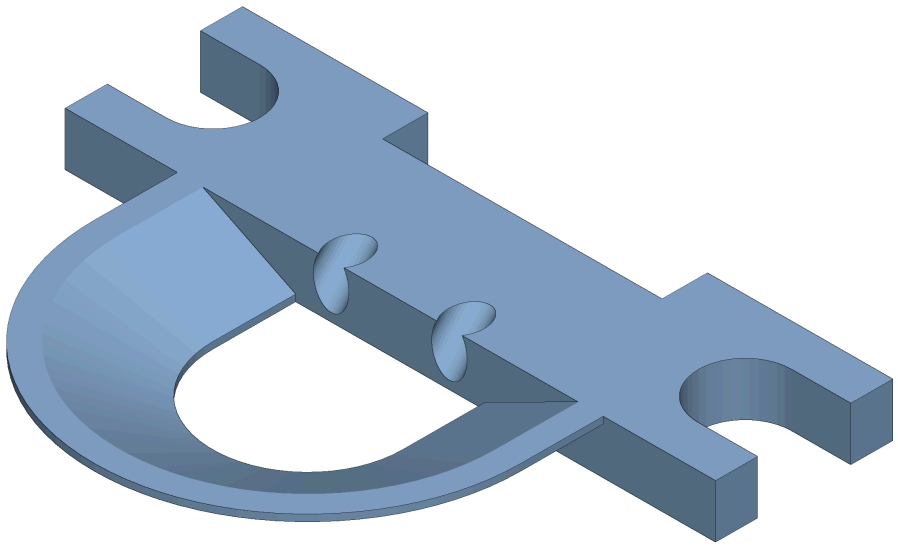
Clear Interlocking Port
Reward port set into the arena floor. Printed clear so the backlit floor shines through and the camera below can capture licks. Locks into the interlocking base.



Domed Port
Dome over the port keeps mice from walking across the ports, and shapes the light for the base camera so licks are captured more reliably.

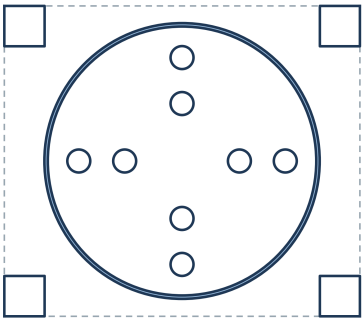


Interlocking Ports Base
Glued to the arena floor. Holes align each port with the floor openings, and built-in hooks route and hold the tubing.



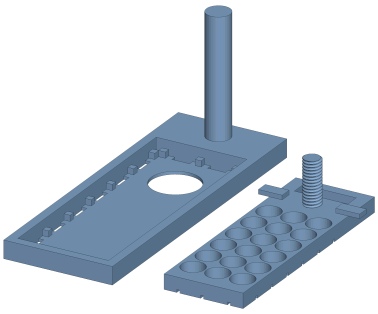
Circular Head Mount

Holds mice head-fixed on the rig while encircling the craniotomy site, which makes surgery easier. The ring forms a well that keeps saline over the skull between behavior sessions, improving animal welfare. Includes two fiber-optic ports and a grounding pin.



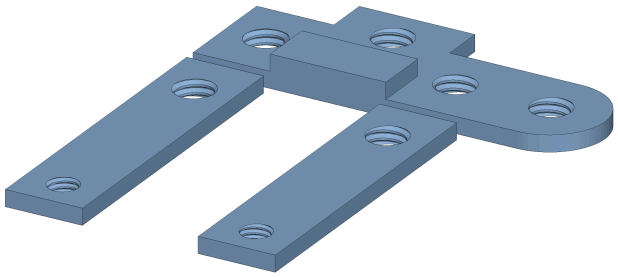
Laser-Cut Arena Floor

Six revisions. An engraved 24-inch ring (shaded) seats the acrylic arena wall, and eight port openings in a cross pattern line up with the interlocking bases.



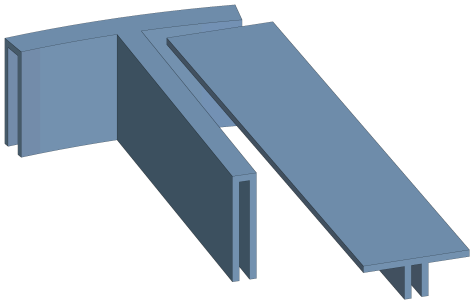
Parametric Plating Aid

Adjustable wells for plating different brain slices onto microscope slides. It made my plating faster and cleaner; I now out-plate a postdoc with over a decade of experience.



Surgery Mount Holder

Custom-fit to our surgery rig. It holds the head mount in place while it's attached, and several attachment options keep the skull visible during placement.



Arena Wall Holders

Hold the acrylic arena walls rigid so they don't bend or give way, even with mice climbing and pushing on them.